Simulation Results and Analysis

Overview of Simulation Study

The rocket design was developed and evaluated using **OpenRocket**, an open-source rocket simulation software widely used for aerodynamic analysis, stability evaluation, and flight performance prediction. Multiple simulations were conducted using different motor configurations to study the influence of propulsion systems on altitude, velocity, acceleration, and overall flight characteristics.

The objective of the simulation was to:

- Validate rocket stability before launch.
- Compare performance under different motor configurations.
- Estimate maximum altitude (apogee).
- Analyze velocity and acceleration profiles.
- Determine recovery system deployment characteristics.
- Evaluate overall flight safety and efficiency.

The simulations provide a virtual environment for assessing rocket performance before physical manufacturing and testing, reducing design risks and improving reliability.

Rocket Design Configuration

The rocket consists of several major components that contribute to its aerodynamic performance and flight stability.

Nose Cone

A streamlined ogive-shaped nose cone was selected to minimize aerodynamic drag during ascent and improve airflow around the rocket body.

Body Tube

The cylindrical body serves as the primary structural component and houses the recovery system, payload section, and other internal components.

Fins

The rocket utilizes a three-fin configuration located near the aft section. These fins generate restoring aerodynamic forces that maintain stability throughout the flight trajectory.

Motor Mount

A rear-mounted motor housing accommodates different motor classes used during simulation testing.

Recovery System

A parachute-based recovery system was incorporated to ensure safe descent and minimize landing impact forces.

Stability Analysis

One of the most critical parameters in rocket design is aerodynamic stability. OpenRocket calculates stability using the relationship between the Center of Gravity (CG) and the Center of Pressure (CP).

For a rocket to maintain stable flight, the Center of Gravity must remain ahead of the Center of Pressure. The simulation results show that the rocket maintained a positive stability margin across all tested configurations.

A stable rocket exhibits the following characteristics:

- Maintains its intended flight path.
- Reduces oscillations during ascent.
- Minimizes the possibility of tumbling.
- Improves aerodynamic efficiency.

The CG-CP relationship observed in the simulations confirms that the rocket design is aerodynamically stable and suitable for flight operations.

Motor Performance Comparison

Several motor configurations were analyzed to evaluate their influence on flight performance.

H128W-6 Motor

Parameter	Value
Apogee	733–761 m
Maximum Velocity	256 m/s
Maximum Acceleration	326–328 m/s²
Flight Time	134–140 s

This motor provided moderate altitude performance while maintaining manageable acceleration levels.

H170-10A Motor

Parameter	Value
Apogee	1011 m
Maximum Velocity	282 m/s
Maximum Acceleration	424 m/s²
Flight Time	142 s

The H170 motor produced significantly higher altitude and acceleration compared to the H128 configuration.

J350W-14A Motor

Parameter	Value
Apogee	1523–1587 m
Maximum Velocity	592–598 m/s
Maximum Acceleration	364–592 m/s²
Flight Time	113–185 s

The J350 motor achieved the highest overall performance among all tested configurations.

Altitude Analysis

Altitude is one of the most important indicators of rocket efficiency. The simulation results demonstrate a clear improvement in apogee with increasing motor thrust.

Motor	Maximum Apogee
H128W-6	761 m
H170-10A	1011 m
J350W-14A	1587 m

The J350 motor achieved approximately 108% greater altitude than the H128 configuration.

Higher thrust enables the rocket to accelerate more rapidly during powered flight, reducing gravity losses and increasing the energy available for upward motion. Consequently, the rocket reaches significantly higher altitudes before entering the coast phase.

The simulation results demonstrate that propulsion system selection plays a major role in determining maximum altitude and mission capability.

Velocity Analysis

Maximum velocity directly affects the aerodynamic loads experienced during flight.

Motor	Maximum Velocity
H128W-6	256 m/s
H170-10A	282 m/s
J350W-14A	598 m/s

The J350 configuration approached approximately Mach 1.76, indicating near-supersonic flight performance.

Higher velocities increase aerodynamic drag and structural loading. Therefore, rockets operating at these speeds require careful aerodynamic design and structural validation. Despite the significant increase in velocity, the rocket remained stable throughout the simulated flight profile.

These findings confirm that the design can safely accommodate high-speed flight conditions while maintaining acceptable aerodynamic behavior.

Acceleration Analysis

Acceleration determines how rapidly the rocket gains speed after launch and influences both launch stability and flight efficiency.

Motor	Maximum Acceleration
H128W-6	328 m/s²
H170-10A	424 m/s²
J350W-14A	592 m/s²

The J350 motor generated the highest acceleration values among all tested configurations.

Higher acceleration improves launch rail departure speed, reducing the influence of wind disturbances during the initial flight phase. Increased acceleration also contributes to higher altitude capability by providing greater kinetic energy during powered ascent.

However, higher acceleration produces larger structural loads that must be considered during rocket design and material selection. The simulation results indicate that the rocket structure remained within acceptable operating conditions throughout the acceleration phase.

Recovery System Evaluation

A reliable recovery system is essential for ensuring safe rocket retrieval and minimizing structural damage after flight.

Deployment Velocity

Configuration	Deployment Velocity
H128W-6	0.32 m/s
H170-10A	0.22 m/s

Configuration	Deployment Velocity
J350W-14A	0.52 m/s

The low deployment velocities indicate appropriate delay timing and successful parachute deployment conditions.

Landing Velocity

Configuration	Ground Impact Velocity
H128W-6	6.1 m/s
H170-10A	6.7 m/s
J350W-14A	7.2 m/s

These values fall within acceptable limits for model rocket recovery systems and suggest a low probability of landing-related structural damage.

The recovery analysis confirms that the parachute system is capable of providing a controlled descent and safe landing across all motor configurations.

Flight Duration Analysis

The total flight duration includes powered ascent, coast phase, parachute deployment, and descent.

Motor	Flight Time
H128W-6	140 s
H170-10A	142 s
J350W-14A	113–185 s

Higher altitude flights generally resulted in longer overall flight durations because the rocket spent more time descending under parachute recovery.

The extended flight durations observed for high-performance motor configurations demonstrate the effectiveness of the recovery system in maintaining safe descent rates while allowing the rocket to achieve significant altitude gains.

Summary of Results

The OpenRocket simulations successfully validated the rocket design and demonstrated the impact of motor selection on flight performance.

The analysis revealed stable aerodynamic behavior across all tested configurations, with a positive stability margin maintained throughout the flight. The maximum apogee increased from approximately 761 m to 1587 m as motor thrust increased. Maximum velocity approached Mach 1.76 under the J350W-14A configuration, while acceleration increased significantly without compromising stability.

The recovery system performed effectively in all simulations, maintaining low deployment velocities and safe landing speeds. Overall, the J350W-14A motor delivered the highest performance, achieving the greatest altitude, velocity, and acceleration.

The simulation results confirm that the rocket design is capable of stable, high-altitude flight and provides a strong foundation for future prototype development, experimental testing, and advanced aerospace research activities.